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- Review of time-of-flight mass spectrographs
- Versatile application of time-of-flight spectrographs for ions and neutral atoms
- Simultaneous determination of velocity distribution and composition

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Time-of-flight mass spectrographs—From ions to neutral atoms

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Abstract After their introduction to space physics in the mid 1980s time-of-flight (TOF) spectrographs have become a main staple in spaceborne mass spectrometry. They have largely replaced magnetic spectrometers, except when extremely high mass resolution is required to identify complex molecules, for example, in the vicinity of comets or in planetary atmospheres. In combination with electrostatic analyzers and often solid state detectors, TOF spectrographs have become key instruments to diagnose space plasma velocity distributions, mass, and ionic charge composition. With a variety of implementation schemes that also include isochronous electric field configurations, TOF spectrographs can respond to diverse science requirements. This includes a wide range in mass resolution to allow the separation of medium heavy isotopes or to simply provide distributions of the major species, such as H, He, and O, to obtain information on source tracers or mass fluxes. With a top-hat analyzer at the front end, or in combination with deflectors for three-axis stabilized spacecraft, the distribution function of ions can be obtained with good time resolution. Most recently, the reach of TOF ion mass spectrographs has been extended to include energetic neutral atoms. After selecting the arrival direction with mechanical collimation, followed by conversion to ions, adapted TOF sensors form a new branch of the spectrograph family tree. We review the requirements, challenges, and implementation schemes for ion and neutral atom spectrographs, including potential directions for the future, while largely avoiding overlap with complementary contributions in this special issue.

1. Introduction

The measurement of particle populations in space electrically charged and neutral alike is undertaken to provide essential information toward several key objectives that are behind typical heliophysics missions, such as (1) origin or source population of locally detected particles, (2) heating of plasmas and neutral gases, (3) transport and interaction of charged and neutral particles, and (4) acceleration of charged particles to high energies.

For the evaluation of particle populations to address these questions it is essential to determine all parameters that characterize each individual particle and then to synthesize the observable distributions separately for all these parameters. The relevant key observables for particles are (1) elemental and sometimes isotopic composition or particle mass A , (2) charge state composition or the ionic charge Q of the particle, and (3) particle speed v , as well as arrival direction θ and Φ

Subdivided into small increments in speed and angle, the particle count rate accumulated over predefined time intervals can be turned into the velocity distribution function $f(v, \theta, \Phi, t)$, including its variation over time t , for ions, electrons, and neutral atoms, separate for individual species. In this paper, we will concentrate on ion and neutral atom distributions. We will treat atoms as particles with charge state $Q=0$. For a plasma physicist, the velocity distribution function f is the key to physical processes that control the interaction in plasmas. Instead of observing directly the particle velocity, particle sensors usually return the kinetic particle energy E along with the arrival direction. The energy E is linked to the speed v with the particle mass A or species. With the geometric factor G the observable particle count rate can be translated into the differential particle flux $\Delta J(E, \theta, \Phi, t)/\Delta E \Delta \Omega$, which is related to the distribution function via

$$\frac{\Delta J(E, \theta, \Phi, t)}{\Delta E \Delta \Omega} = \frac{v^2}{A m_p} f(v, \theta, \Phi, t) \quad (1)$$

where $\Delta \Omega$ is the field of view (FoV) and ΔE the energy resolution of the sensor. By integration over a small

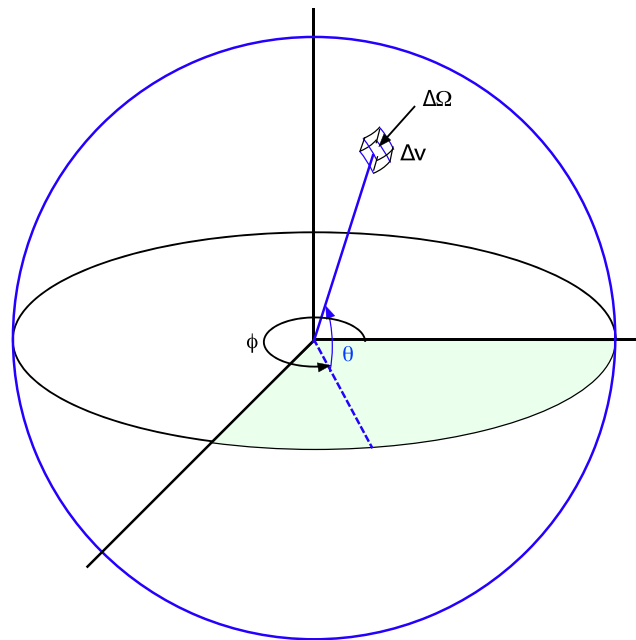


Figure 1. Schematic view of a particle velocity distribution function $f(v, \theta, \Phi)$ and how a particle detector system selects a certain portion as differential particle flux $\Delta J / \Delta E \Delta \Omega$.

phase space volume a particle detector obtains the count rate R in a selected angular sector with center angles θ and Φ and at a selected center energy E , according to

$$R(\theta, \Phi, E, t) = \frac{\Delta J(E, \theta, \Phi, t)}{\Delta E \Delta \Omega} \cdot G \cdot \Delta E \Delta \Omega \quad (2)$$

The situation is depicted schematically in Figure 1.

Any particle counting sensor with energy resolution and a constrained FoV that is either a subdivision of a larger FoV or can be swept across the sky provides the crucial differential particle flux. However, to distinguish and thus accumulate separately particle species according to M and Q requires additional diagnostics. In the early days of space exploration, the combination of electrostatic and magnetic deflection was employed [e.g., *Coplan et al.*, 1978; *Shelley et al.*, 1978] to provide mass resolution for bulk plasmas up to about 10 keV/ Q . This technique is still very important for very high mass resolution in connection with in situ sampling of planetary, lunar, and cometary atmosphere environments [e.g., *Balsiger et al.*, 2007]. For energetic particle populations, such as solar, anomalous, and galactic cosmic rays, particle telescopes are used that obtain the energy loss ΔE in a thin detector and measure the residual energy E_R in a thick detector, where the particles are stopped, to obtain species resolution. These telescopes make use of the nuclear charge Z and mass A dependent specific energy loss in bulk material. When including gas proportional counters as ΔE elements [Hovestadt and Vollmer, 1971; Hovestadt et al., 1978], this technique covers energies down to <100 keV/nucleon. The two aforementioned methods still left a substantial gap in energy coverage, in particular in the crucial suprathermal regime where ions are accelerated out of their source populations, until the arrival of time-of-flight (TOF) mass spectrographs in the 1980s [Gloeckler and Hsieh, 1979; Gloeckler, 1990]. Wüest [1998] provided a review in the first symposium on space plasma instrumentation.

In this paper, we will give a current overview of TOF mass spectrometry for ion and neutral atom populations with simultaneous diagnostics of their distribution functions. It is largely based on a presentation at the conference on Measurement Techniques in Solar and Space Physics and avoids substantial overlap with concurrent presentations by Gilbert, McFadden, Paschalidis, and Zurbuchen on TOF spectrometry and by Barabash and Funsten on neutral atom imaging. Where related sensors are addressed in papers for this special issue, they are cross referenced in the appropriate section. Section 2 will provide a brief introduction to the TOF method, its strengths and challenges. TOF spectrographs for suprathermal and energetic particle populations will be covered in section 3, those for thermal plasma populations and the solar wind in section 4, high-resolution mass

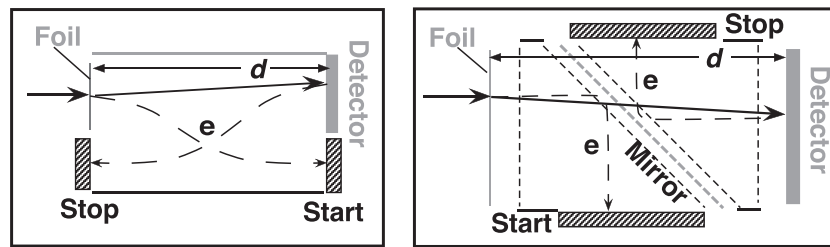


Figure 2. Schematic view of generic TOF sensors. Particles enter through a thin C-foil from the left and traverse the TOF path d to a detector on the right. Secondary electrons are guided by electric fields to start and stop MCPs. Structure in black is held at 1–2 kV positive relative to structure in grey. The detector on the right may be an MCP or a solid state detector to measure energy (see section 3). (left) Electrons cross before reaching the start and stop MCP. Grids after the foil and in front of the detector at positive potential are often used to accelerate the electrons more quickly to reduce dispersion. (right) Electrons are reflected by a dual electrostatic mirror, which also facilitates 2-D position sensing.

spectrographs in section 5, and neutral atom imagers with mass resolution in section 6. Section 7 contains a brief summary and indicates where TOF spectrographs may be key in the near future.

2. Brief Introduction to the Time-of-Flight Method

Compared with simple particle detectors that typically count single signals and thus are vulnerable to any noise or background count, a TOF detector, by means of its method, is a genuine coincidence detection system, which provides intrinsic noise and background suppression. As the name implies, for each incoming particle the measurement of the flight time τ between a start and a stop detector across a known distance d is obtained, which requires two independent detections of the same particle. Two generic TOF configurations are shown in Figure 2. The start signal is obtained in most cases from the passage of a thin foil, typically made of carbon, where the particle generates secondary electrons. These are accelerated and steered to an electron multiplying detector (channeltron or microchannel plate, MCP) by means of appropriate electrostatic field configurations. The stop signal may be detected directly or, again, through secondary electrons. In Figure 2 (left), the electrons from the start and stop location are accelerated toward the center of the sensor, cross each other, and are deflected by a transverse field to a start and stop MCP at each end. In Figure 2 (right) a configuration with electrostatic mirrors is shown, which reflect the start and stop electrons to start and stop MCP at opposite sides of the sensor perpendicular to the particle trajectory. While the former configuration contains fewer grids that reduce the sensor transparency and generate more scattered particles, the latter configuration allows to obtain the start and stop position in two dimensions from the signals. The choice depends on the measurement requirements.

The main objective of the TOF section is to determine the speed v of the particles and thus their energy E per mass unit A in front of the entrance foil according to

$$\frac{E_{\text{TOF}}}{A} = \frac{1}{1 - \alpha_1} \frac{\text{amu}}{2} \left(\frac{d}{\tau} \right)^2 \quad (3)$$

amu (1 Dalton) is the atomic mass unit (1/12 of the carbon atom mass), α_1 is a parameter that reflects the energy loss in the entrance foil. For each TOF sensor this energy loss is determined by a combination of simulation [e.g., Ziegler *et al.*, 2015] and calibration. The TOF resolution $\Delta \tau/\tau$ depends mainly on three characteristics.

1. The energy loss in the foil is a statistical process and thus associated with energy straggling that increases with foil thickness, decreasing particle energy, and increasing mass. Large efforts have gone into reducing the foil thickness [e.g., McComas *et al.*, 2004] and/or into boosting the energy of incoming ions with a high postacceleration voltage, as described in sections 4.1 and 4.2.
2. The secondary electrons arrive at each MCP with a time dispersion, which can be minimized by their immediate acceleration with a grid on the positive acceleration potential right in front of the foil and the detector at the end, as shown in the electrostatic mirror configuration in Figure 2 (right). Such grids have also been used in the overcrossing configuration on the left, for example, in Active Magnetospheric Particle Tracer Explorers (AMPTE) SUPrathermal Low Energy Ion Charge Analyzer (SULEICA) [Möbius *et al.*, 1985a].

3. Due to a finite FoV angle range and additional angular scattering in the foil that is causally connected with the energy straggling, the actual flight d' path may be longer than d according to $d' = d/\cos\theta$, where θ is the angle of the flight path relative to the normal direction of the foil plane. The angular range may be reduced by a long and slender TOF system geometry. Alternatively, in the electrostatic mirror configuration, the start and stop position can be measured with MCP position sensing, providing the information to correct the flight path d' .

Typically, energy straggling (1) dominates the overall TOF resolution $\Delta\tau/\tau$ at low energies because it becomes rather strong. For high-energy or high-speed particles, the electron time dispersion (2) along with the timing accuracy of the electronics dominates, because this is a fixed amount for a given instrument concept, while the TOF τ decreases with speed. A finite angular range (3) leads to a fixed ratio $\Delta\tau/\tau$ and thus a constant contribution to the overall resolution.

As is evident, to measure the TOF τ both the start and the stop signal are needed, thus providing a natural double coincidence, with intrinsic background and noise suppression by orders of magnitude compared with single signal detection techniques. Assuming a noise or background detection rate R_N with a single MCP the statistical background contribution R_{BG} to the TOF rate that competes with a TOF signal with timing resolution $\Delta\tau$ can be estimated as

$$R_{BG} = R_N^2 \cdot \Delta\tau \quad (4)$$

For a TOF resolution of a few nanoseconds even noise rates of a few kHz are reduced to rates well below 0.1 s^{-1} . However, the presence of a majority species, typically H, can provide challenges to the detection of minor species with abundances many orders of magnitude below that of H. The detection efficiency for secondary electrons is always lower than 100%. Therefore, particles whose start signal was not detected but their stop signal was may produce chance coincidence events with other ions that had a start signal. Assuming an incoming rate R_{IN} of the majority species, the resulting chance coincidence background rate R_{BG} is

$$R_{BG} = (1 - \mu_{\text{Start}}) \cdot \mu_{\text{Stop}} \cdot R_{IN}^2 \cdot \Delta\tau \quad (4a)$$

where μ_{Start} and μ_{Stop} are the start and stop efficiencies for the majority species. While equation (4a) still contains the coincidence related background suppression term $R_{IN} \Delta\tau$, the background rate now increases with the square of the incoming rate.

In general, a TOF sensor provides a speed or energy per mass measurement, while simultaneously giving noise and background suppression. This detection technique works equally well for incoming ions or neutral atoms. To determine the mass number A and/or the ionic charge state Q (where a neutral atom has $Q=0$) a TOF system needs to be combined with other measurements that provide the total energy E and/or the energy per ionic charge E/Q . From the combination of different energy analysis techniques with a TOF subsystem a family of TOF sensors emerges, each of whose members can be optimized for a suitable range of specific applications tailored to the goals of a space mission.

3. Suprathermal/Energetic TOF Spectrographs

As pointed out in section 1, the suprathermal energy range had been challenging to cover with species-sensitive particle sensors before the advent of the TOF method. Indeed, the first TOF spectrographs were implemented for this energy range on satellites in the 1980s. The first mission to contain several TOF spectrographs was the Active Magnetospheric Particle Tracer Explorers (AMPTE) mission [Bryant *et al.*, 1985], with three such instruments, the Medium Energy Particle Analyzer (MEPA) [McEntire *et al.*, 1985] and the Charge Energy Mass (CHEM) [Gloeckler *et al.*, 1985] sensor on the Charge Composition Explorer, and the SUPrathermal Low Energy Ion Charge Analyzer (SULEICA) [Möbius *et al.*, 1985a] on the Ion Release Module (IRM).

The MEPA sensor is an example for the combination of a TOF system with solid state detectors (SSD) to determine separately the residual energy E_{Res} , i.e., the energy of E of the incoming article reduced by the fraction α_1 lost in the entrance foil and the fraction α_2 in the contact layer of the SSD. This detector, where the particles are stopped and deposit their energy, is included in the schematics in Figure 2.

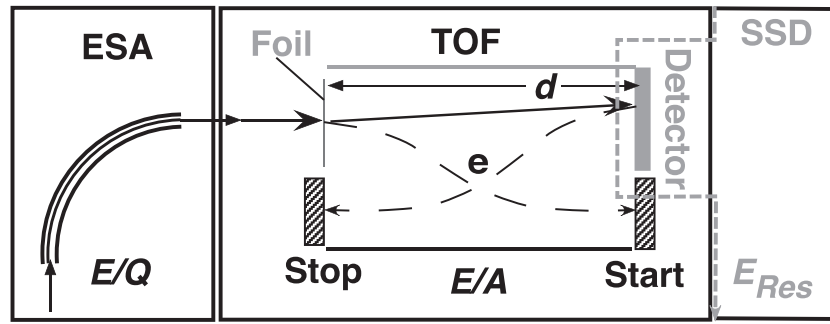


Figure 3. Schematic view of an ESA-TOF combination with an SSD. The ESA serves as E/Q filter. The TOF spectrograph provides the energy per mass E/A according to equation (3). The SSD measures the residual energy E_{Res} and thus allows a separate determination of E , A , and Q .

Having determined the energy per mass E/A in equation (3), we now obtain the atomic mass A of the particle according to

$$A = \frac{E_{Res}}{(1 - \alpha_2) \cdot E_{TOF}/A} \quad (5)$$

As α_1 the energy loss α_2 is energy and species dependent and is determined by calibration with known ion beams at an accelerator. It is evident that the total energy E can also be determined.

The MEPA sensor was implemented with the electrostatic mirror. To cope with the high intensity of penetrating radiation in the radiation belts and ring current, MEPA included a second foil at the center grid of the electrostatic mirror. From here the secondary electrons were accelerated to a third MCP in the corner of the sensor to provide another independent coincidence measurement, which substantially reduced the background. Other examples of this sensor type are the Low Energy Ion Composition Analyzer (LICA) on the Solar Anomalous and Magnetospheric Particle Explorer (SAMPEX) [Mason *et al.*, 1990, 1993], the Ultra Low Energy Isotope Spectrometer (ULEIS) on the Advanced Composition Explorer (ACE) [Mason *et al.*, 1998], and the Suprathermal Ion Telescope (SIT) on the Solar TErrestrial RElations Observatory (STEREO) [Mason *et al.*, 2008]. To boost TOF resolution and reduce background, ULEIS contains two consecutive electrostatic mirror TOF sections. This sensor allowed abundance measurements of ^3He in small impulsive energetic particle events even below 1%, thus establishing that selective acceleration of ^3He is indeed a hallmark of all impulsive events. Enhanced ^3He abundances were even found in coronal mass ejections indicating reacceleration of remnant impulsive populations [Mason *et al.*, 2005; Desai *et al.*, 2006].

To separately determine E , A , and the ionic charge state Q of incoming ions, the combination with an electrostatic analyzer (ESA) at the sensor entrance is needed. The AMPTE CHEM and SULEICA instruments were the first satellite borne TOF spectrographs to provide that capability. A schematic view of the SULEICA sensor is shown in Figure 3. The ESA acts as energy per charge filter, which passes ions within a narrow band with a specific E/Q and resolution $\Delta E/Q$. The mass per charge ratio is determined as

$$\frac{A}{Q} = \frac{E/Q}{E_{TOF}/A} \quad (6)$$

making use of $E/Q = E_{TOF}/Q$, where E_{TOF} is obtained from equation (3). As a consequence, the full energy range is covered by changing the ESA voltage in logarithmic steps. The number of energy steps n determines the duty cycle of the sensor, and its effective geometric factor is reduced by the factor n . We will turn to potential improvements in section 4.2.

Typically, the ions are first separated according to A/Q with the combined resolution of the ESA and the TOF section according to

$$\frac{\Delta A/Q}{A/Q} = \sqrt{\left(\frac{\Delta E/Q}{E/Q}\right)^2 + 2\left(\frac{\Delta \tau}{\tau}\right)^2} \quad (6a)$$

The ESA resolution is constant for a given analyzer, whereas the TOF resolution varies with species and energy. The resolution in mass A is a combination of the TOF resolution and the energy resolution of the SSD.

$$\frac{\Delta A}{A} = \sqrt{\left(\frac{\Delta E_{\text{Res}}}{E_{\text{Res}}}\right)^2 + 2\left(\frac{\Delta \tau}{\tau}\right)^2} \quad (6b)$$

Equations (6a) and (6b) are good approximations as long as the contributing resolution ratios are small compared to one. Both the TOF and the energy resolution increase rapidly for low particle energies. Because the energy range of CHEM extends into the bulk plasmas of the magnetosphere and the solar wind, this sensor includes postacceleration to boost the ion energy for the TOF and energy measurement [Gloeckler *et al.*, 1985], as discussed in section 4.3.

The A/Q resolution of SULEICA alone was key for the discovery of interstellar He^+ pickup ions [Möbius *et al.*, 1985b] in conjunction with detecting Li^+ [Möbius *et al.*, 1986] from the cloud experiments in the solar wind conducted with the AMPTE IRM satellite [Haerendel *et al.*, 1985]. The combination of A/Q resolution and total energy measurement allowed the unambiguous identification of energetic NO^+ and O_2^+ ions in a poststorm ring current [Klecker *et al.*, 1986], because molecular ions typically break up when passing the TOF entrance foil. In this way, suprathermal TOF mass spectrographs opened the door for many studies, which were not possible before. Other examples of this sensor type are WIND STICS [Gloeckler *et al.*, 1995] and SOHO STOF [Hovestadt *et al.*, 1995].

4. Low-Energy TOF Spectrographs for Bulk Plasmas

Until the 1980s magnetic mass spectrometers dominated the instrumentation in the thermal and bulk plasma regime. However, their limitation to energies below about 15 keV/ Q , along with the difficulty to provide a wide angular FoV and their relatively low geometric factor, provided formidable challenges to measure three-dimensional distribution functions with mass and charge resolution and a reasonably high time resolution.

4.1. Distribution Function Analyzers With A/Q Resolution

The realization that only small contributions of a few percent of O^+ ions to magnetospheric plasmas or 10% of He^{2+} ions to the solar wind would have a substantial contribution to the momentum balance in physical processes, such as reconnection and bulk acceleration led to an alternate approach that includes the TOF method. Thus, for Cluster and simultaneously for FAST and Equator-S a sensor concept was developed based on a marriage between a top-hat ESA, as used successfully in the 3-D Plasma instrument [Paschmann *et al.*, 1985], and the TOF section of the SULEICA sensor [Möbius *et al.*, 1985a], both on AMPTE IRM. For the new sensor, the TOF section was expanded to a fully rotationally symmetric cylindrical geometry [Möbius *et al.*, 1998a; Rème *et al.*, 1997; Klumpar *et al.*, 2001]. Because the most important ion species in the magnetosphere (H^+ , He^{2+} , He^+ , O^{2+} , and O^+) are separated in A/Q by subsequent factors of 2 an implementation could be chosen that consists solely of an ESA and a TOF section to determine E/Q and E/A , albeit with postacceleration of the ions between ESA and TOF section with a negative voltage up to -20 kV.

Figure 4 includes a cross sectional and a top view of the COMposition and Distribution Function (CODIF) sensor. The top-hat ESA, for which the lower part of Figure 4 represents a radial cut of the fully rotationally symmetric configuration, allows instantaneous coverage up to 360° in elevation angle θ and typically $\Delta\Phi = 6\text{--}15^\circ$ in azimuth. Each spin is subdivided into 32 spin angle (Φ) sectors during each of which the ESA voltage is swept through its full range to provide energy coverage. For Cluster CODIF and the Equator-S Ion Composition sensor (ESIC), each sweep is subdivided into 128 logarithmically spaced steps, while FAST Time-of-Flight Energy, Angle, Mass Spectrograph (TEAMS) uses 192 steps. The Cluster CODIF and the Equator-S ESIC sensors have a split FoV with two 180° wide apertures and geometric factors that differ by about $\times 50$ to increase the dynamic range and to allow accommodation of a retarding potential analyzer for the lowest energies [Rème *et al.*, 1997]. FAST TEAMS makes use of the full 360° aperture with identical geometric factor and thus obtains a full distribution function within one half spin. The relevant E/Q for the TOF section includes the postacceleration voltage U_{PAC} , thus modifying equation (6) to

$$\frac{A}{Q} = \frac{E/Q + U_{\text{PAC}}}{E_{\text{TOF}}/A} \quad (7)$$

The design challenges and choices for this sensor were described by Möbius *et al.* [1998a] in connection with the previous space instrumentation symposium and in the instrument papers and thus shall not be repeated

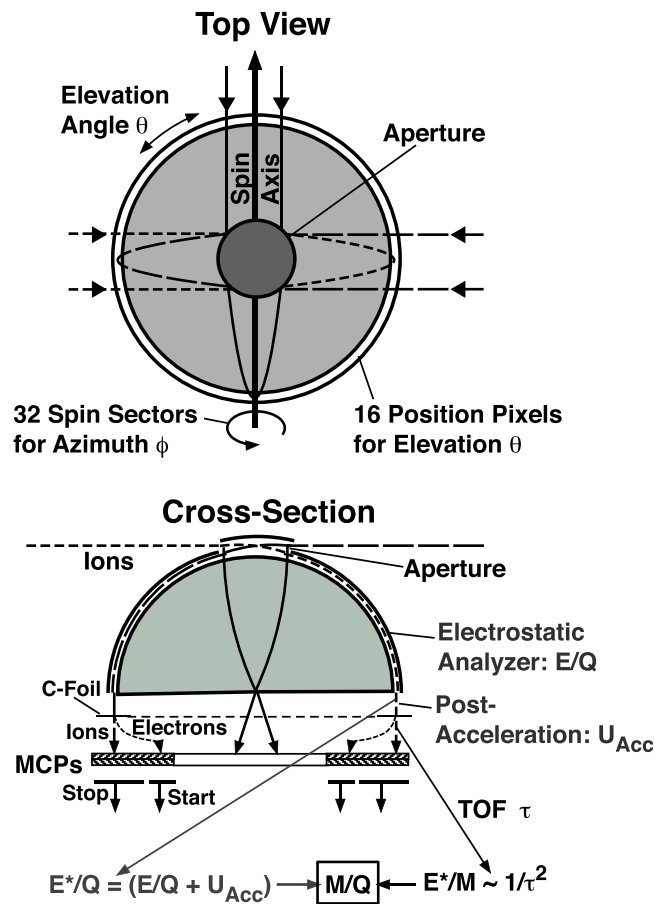


Figure 4. Top view and cross-sectional view of the CODIF sensor. The top-hat ESA provides imaging of elevation angle θ to a position at the exit, which the start electrons map to 16 sectors of the start MCP. Spin covers azimuth Φ because the sensor symmetry axis is oriented at 90° relative to the spin axis.

boring charge states of, for example, C and O, a combined resolution in A and Q of better than $\Delta A, \Delta Q \leq 1$ is needed. To achieve such resolution postacceleration voltages in excess of 20 kV [Gloeckler *et al.*, 1992], ultra-thin C-foils [Funsten *et al.*, 1994], and SSDs with very good energy resolution and thin dead layers are needed. A typical instrument for solar wind composition measurements is Ulysses and ACE Solar Wind Ion Composition Spectrometer (SWICS) [Gloeckler *et al.*, 1992, 1998]. The sensor is similar to the one in Figure 3, but with postacceleration between the ESA and the TOF section. Now the entire TOF section, including SSD and analog electronics, is at the postacceleration potential. Like AMPTE SULEICA and Cluster CODIF, SWICS takes advantage of the spinning Ulysses and ACE spacecraft to obtain angular distributions. With the spin axis pointing close to the Sun, the SWICS FoV of 70° in the direction perpendicular to the spin plane guarantees almost continuous coverage of the solar wind and extensive coverage of pickup ions.

The other two solar wind composition TOF spectrographs, the CTOF sensor of the Charge, Element and Isotope Analysis System for (CELIAS) on the Solar and Heliospheric Observatory (SOHO) [Hovestadt *et al.*, 1995] and the STEREO Plasma and Suprathermal Ion Composition (PLASTIC) sensor [Galvin *et al.*, 2008], were implemented on three-axis stabilized spacecraft that were optimized for Sun-viewing optical and EUV cameras. With the large collecting power of its 180° hemispherical electrostatic analyzer situated between two quadrupole lenses and uninterrupted coverage of the solar wind, CTOF provided the first high time resolution measurements of solar wind composition. Like its predecessor CTOF, STEREO PLASTIC provides uninterrupted and high time resolution for solar wind ion measurements, but PLASTIC also provides the first full directional information for solar wind heavy ions. The entrance system for PLASTIC [Blush *et al.*, 2005] is a 360° top-hat electrostatic analyzer that incorporates three different geometrical factors. In the sunward

here. The sensors on Cluster, FAST, and Equator-S have led to numerous studies that would have been impossible without the fast coverage of the distribution function and the species identification, for example: differential acceleration of ions in the auroral region [Lund *et al.*, 1998; Möbius *et al.*, 1998b], acceleration of He^+ by electromagnetic ion cyclotron waves in the inner magnetosphere [Mouikis *et al.*, 2002], the importance of the cusp as a source for O^+ in the magnetotail [Kistler *et al.*, 2010], and the role that O^+ plays in magnetic reconnection in the magnetotail [Liu *et al.*, 2015] and at the magnetopause [Wang *et al.*, 2015].

4.2. TOF Sensors With Postacceleration That Provide Charge State

While the most abundant ion species in the Earth's magnetosphere as well as interstellar pickup ions can be distinguished unambiguously with an ESA-TOF combination, a meaningful composition analysis of the solar wind and related suprathermal particles requires a three-dimensional analysis [E/Q , E/A , E], including residual energy measurement with an SSD. To distinguish between neigh-

direction, two sets of electrostatic deflectors are also used in order to provide 40° coverage in the out-of-ecliptic direction. For solar wind ions, the largest geometric factor is reduced automatically by electrostatically switching to the smallest aperture during each ESA stepping cycle, when high He^{2+} and H^+ fluxes are encountered [Galvin *et al.*, 2008]. In this manner, reliable measurements over the entire dynamic range of suprathermal and solar wind ion populations can be obtained in each 1 min cycle. In contrast, on other missions, an additional sensor is included in the payload for the purpose of measuring solar wind protons and alphas. Like SWICS and CTOF, the PLASTIC TOF section incorporates SSDs to allow full mass and charge measurements.

Among others, Ulysses and ACE SWICS have allowed great advances toward the composition, velocity distribution, and spatial distribution of pickup ions [Gloeckler and Geiss, 1998; Gloeckler *et al.*, 2004; Chen *et al.*, 2013]. They have demonstrated distinct mass and charge composition differences between fast and slow solar wind and their variations with solar activity [von Steiger *et al.*, 2000] as well as great insight into coronal mass ejection related solar wind [Lepri *et al.*, 2001] and their likely source populations. The unique high angular resolution of STEREO PLASTIC has recently allowed the unambiguous demonstration that interstellar He^+ pickup ions maintain a narrow toroidal velocity distribution from the original injection for quite a while during their transport with the solar wind [Drews *et al.*, 2015]. The temporal resolution of SOHO CTOF and STEREO PLASTIC (e.g., 5–10 min for an iron spectrum) has enabled distinguishing differences in the magnetic field and ion compositional boundaries of solar wind structures [e.g., Aellig *et al.*, 1998; Galvin *et al.*, 2013; Winslow *et al.*, 2016].

4.3. Efforts to Improve Resolution, Temporal Cadence, and Reduce Sensor Size

One key capability of the various plasma TOF spectrographs is their ability to separate individual species and charge states. However, to resolve unambiguously the charge states of neighboring elements, the minimum energy before entry of the TOF section has to be boosted by postacceleration to relatively high voltages and/or the energy resolution of SSDs has to be improved, in particular for low-energy ions. However, providing isotope resolution requires a different TOF section concept that sacrifices ionic charge resolution and will be discussed in section 5. An extremely challenging method to improve the overall resolution, albeit with some recent success in the laboratory, is to push the postacceleration voltage to 50–100 kV, as presented by Gilbert *et al.* [2016], which would lead to energy, mass, and charge resolution comparable to a suprathermal or energetic particle sensor. Alternatively, the use of an ultrathin window and low-noise SSDs in the form of avalanche photodiodes or novel diamond detectors [Ogasawara *et al.*, 2015] is being explored, which may also allow the combination of energy measurement and stop detection in the same detector and thus could reduce sensor complexity.

The increased complexity of instruments like PLASTIC to provide angular coverage and resolution as well as simultaneous optimization for full mass and charge resolution of the solar wind proper, along with the dual need to increase time resolution for reconnection studies and to reduce the size and complexity of sensors for smaller spacecraft missions, has prompted the search for alternative sensor concepts. One key obstacle to improving time resolution and reducing sensor size, while maintaining a high effective geometric factor, is the intrinsic requirement to step through the ESA voltage. Including the scan in at least one angle, either via the spacecraft spin or with a deflector system, this amounts to nested scans in two dimensions of velocity space. This greatly impacts both time resolution and effective geometric factor because of the low measurement duty cycle. Therefore, several attempts have been and are being made to eliminate one of the scans, either in energy or angle by a scheme that allows simultaneous coverage.

The MESSENGER (MERcury Surface, Space ENVironment, GEOchemistry, and Ranging) Fast Imaging Plasma Spectrometer sensor is a successful attempt to reduce the sensor size and eliminate the need of angular scanning by combining a wide-open ESA with a collimator and position sensing at the interface between ESA and TOF section to obtain the arrival angle for a selected E/Q range [Andrews *et al.*, 2007; Zurbuchen and Gershman, 2016]. A somewhat different and conceptually simplified way to obtain a simultaneous image of the velocity distribution with a FoV of almost 2π has been presented by Skoug *et al.* [2016]. Here the ESA is the region between a planar entrance plate with a center pinhole aperture and a filter plate at the exit, which only passes ions of a predetermined ratio $E/(Q U_{\text{ESA}})$. Yet ions that arrive from a range of elevation angles θ are identified simultaneously, where θ translates into a unique radial position $r(\theta)$. A challenge of this design is obtaining a reasonable energy resolution for particles that arrive at elevation angles close the center axis of the sensor

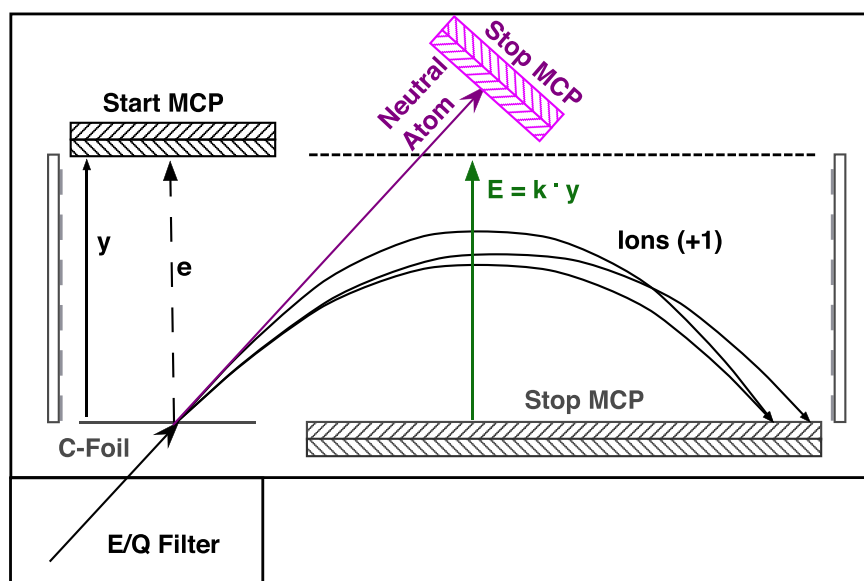


Figure 5. Schematic view of a reflecting TOF spectrograph. In the linearly increasing electric field ions that leave the start foil perform half a harmonic oscillation, whose TOF τ only depends on M/Q , before reaching the stop MCP. Particles that leave the foil as neutral atoms can be analyzed in a linear TOF section (in purple). To obtain the original energy, an E/Q filter may be added to the sensor entrance.

because the deflection field is almost in the direction of motion. To improve this resolution, an additional center cone at the potential of the collimator plate is introduced, which provides a radial deflection field component.

Another attempt to improve the temporal cadence by providing partial imaging of a combination of elevation angles and energy per charge has been made with an ESA curvature that depends on the azimuth angle of the instrument aperture. The Advanced Mass and Ionic Charge Composition Experiment provides simultaneous coverage over a range of energies, thus reducing the number of energy steps by about a factor of 10 [Allegrini *et al.*, 2009, 2016]. With a “flower-shaped” ESA that has a wide opening at the entrance and exit, but is narrow in its middle, the Suprathermal Ion Monitor provides simultaneous coverage over a range in energy and imaging of about 40° in elevation angle [Desai *et al.*, 2015; Desai *et al.*, 2016]. Elevation angle and E/Q are obtained by position sensing at the detector.

5. High-Resolution TOF Spectrographs

Linear TOF spectrographs are limited in their resolution by the spread in energy E and angle θ due to scattering in the C-foil according to $\Delta\tau/\tau \propto \sqrt{\Delta E/E}$ and $\Delta\tau/\tau \propto \sqrt{1 - \cos 2\Delta\theta}$. Uncertainties in the TOF electronics can be held at typically <0.5 ns and are mostly negligible. As discussed already in section 2, the energy and angle scattering are inherent in the design of linear TOF spectrographs and, as discussed in section 4, can only be mitigated within limits by the choice of very thin C-foils and/or high postacceleration voltages. Because the TOF resolution is directly linked to the M/Q and M resolution of the instrument (equations ((3)a) and ((3)b)) separation of neighboring elements and isotopes calls for a different instrument concept. Alternate TOF spectrometry concepts that avoid a foil as start sensing element are in use for specific applications. A rotating electric field, which ties the start timing to the phase of the field, provides a successful alternative to magnetic gas analyzers but without access to the velocity distribution function. An example is the STart from a ROTating Field Optics (STROFIO) sensor on Bepi Colombo [Orsini *et al.*, 2010]. Gated TOF spectrographs that derive the timing by short opening pulses of the system are useful to diagnose high bulk ion fluxes when a small duty cycle of the sensor due to the short opening pulse compared with the TOF window can be afforded [Orsini *et al.*, 2010]. Here we concentrate on isochronous TOF geometries.

5.1. Isochronous TOF Spectrographs

To eliminate the uncertainties related to the energy and angle straggling, a TOF configuration has been implemented, in which the ions follow a ballistic trajectory in a retarding electric field whose strength increases linearly with distance from the common plane of the entrance foil and stop detector as shown in Figure 5 [e.g., Managadze, 1986]. The linearly increasing electric field $E(y) = k \cdot y$ turns the TOF sensor into a harmonic oscillator, and τ becomes half the oscillator period, which solely depends on the ion mass per charge A/Q . The majority of ions in the energy range 1 to 30 keV/e leave the entrance foil of the TOF system as neutral atoms or as singly charged ions [Gonin et al., 1992]. For the ions, the τ obeys

$$\tau = \frac{\pi}{\omega} = \pi \sqrt{\frac{A \cdot \text{amu}}{ek}} \quad (8)$$

which requires a retarding potential of ≈ 20 kV for 45° trajectories. Neither initial energy nor incoming angles affect the resulting TOF τ in the reflecting section, which solely depends on A for singly charged ions, thus making the mass resolution also independent of the ESA E/Q resolution. For an earlier review, see Wüst [1998]. Neutral atoms that leave the TOF section on a straight line can be analyzed in an additional linear TOF section (in purple in Figure 5). To obtain also the energy of the ions, an electrostatic E/Q filter is placed in front of the TOF entrance.

The reflecting electric field, which increases linearly with distance from the plane that contains entrance foil and stop MCP, is key to the harmonic oscillator behavior. Such a field configuration can be implemented with a series of potential shaping electrodes, as in the CAssini Plasma Spectrometer (CAPS) [McComas et al., 1998], or the bounding electrodes are shaped such that they represent the appropriate boundary conditions for the solution of Laplace's equation for a potential that increases with the square of the distance from the entrance plane. This condition has been achieved with a set of V-shaped electrodes for a mostly planar geometry of the ion trajectories in the TOF section, as depicted in Figure 5, for Wind high MAss reSolution Spectrometer (MASS) [Gloeckler et al., 1995] and SOHO CELIAS Mass TOF (MTOF) sensor [Hovestadt et al., 1995]. The Rosetta Orbiter Spectrometer for Ion and Neutral Analysis experiment to study elementary, isotopic, and molecular composition of comet 67P/Churyumov-Gerasimenko not only contains the isochronous Reflecting TOF (RTOF) sensor but also includes a magnetic spectrometer for the mass resolution required to separate neighboring heavy molecular species [Balsiger et al., 2007]. To potentially combine a reflecting TOF section with a top-hat ESA that can obtain instantaneously a 360° slice of the ion velocity distribution, a cylindrically symmetric potential configuration is used in the CYLindrical Mass Analyzing Sensor (CYLMAS) concept [Möbius et al., 1990; Gubler et al., 1995].

5.2. Solar Wind Isotope Observation

WIND MASS [Gloeckler et al., 1995] and SOHO CELIAS MTOF [Hovestadt et al., 1995] have indeed provided great insight into the solar wind isotope composition and have expanded on the initial measurements taken with solar wind collector foils during the Apollo missions on the Moon's surface [Geiss et al., 1972]. The isotope composition of the solar wind is the best proxy that we can currently obtain on the Sun's isotope composition, which in turn is an archive of the element forming processes in the galaxy [Geiss and Reeves, 1972] and of the fractionation processes during the solar system formation [Marti and Bochsler, 2012]. The solar corona and thus the solar wind do not necessarily share an unbiased isotope composition with the Sun's interior [Bochsler, 2001]. Thus, any variations of the isotope ratios with solar wind type and speed will help to unravel fractionation processes between the Sun and the solar wind. To this end, measurements with isochronous TOF sensors have enabled the differentiation between solar wind types, whereas the foil collector technique largely provided average values. A compilation of solar wind composition, with substantial emphasis on isotopes, may be found in the paper by Wimmer-Schweingruber [2002]. The isochronous TOF mass spectrographs have expanded the number of elements now accessible to solar wind isotope observations, including elements up to the mass of iron.

Wimmer-Schweingruber [2002] compared solar wind isotope ratios, among them for Ne and Ar, as obtained with SOHO MTOF [Kallenbach et al., 1997] and with the Apollo foil collectors [Geiss et al., 1972]. All values agree within their observational uncertainties, which serve as a validation of both techniques. Furthermore, the Mg isotope ratios [Bochsler et al., 1997; Wimmer-Schweingruber et al., 1999] show a significant difference between fast and slow solar wind. The values for coronal mass ejections appear to agree within their uncertainties with those of the slow solar wind. This isotope fraction appears to be connected to a dependence of the Mg isotope

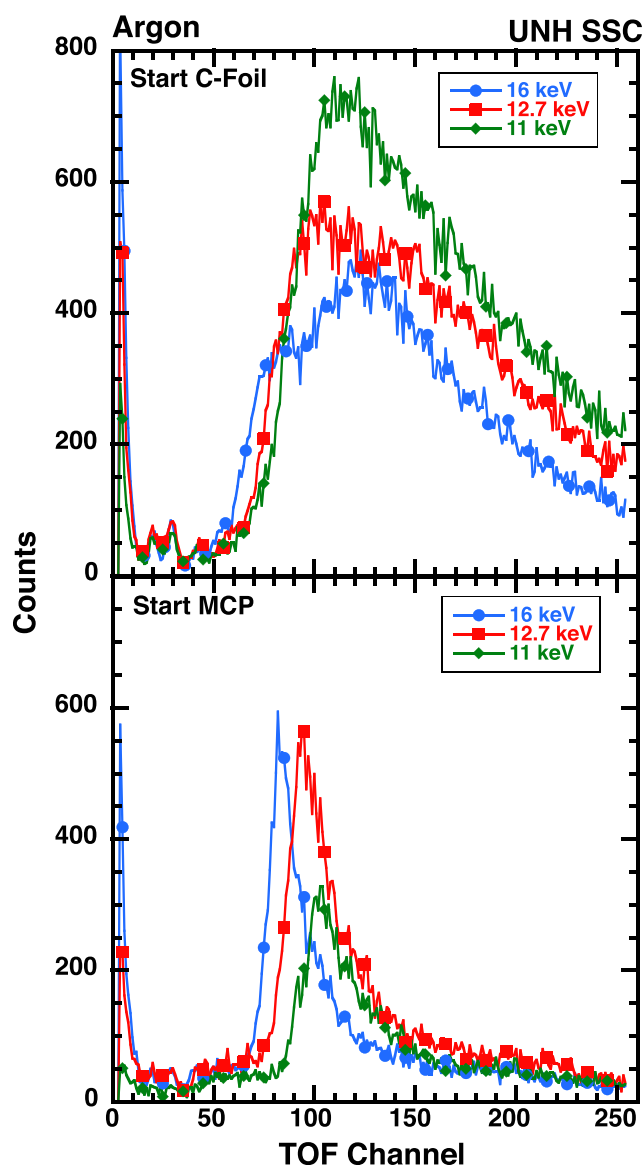


Figure 6. TOF spectra of 11, 12.7, and 16 keV Ar using a $\approx 3.5 \mu\text{g}/\text{cm}^2$ (top) C-foil and a single-stage (bottom) MCP as start window.

single-stage MCP [Devoto *et al.*, 2008]. Instead of penetrating the material of a thin foil, the ions bounce off the surface inside the electron multiplication channels of the MCP at glancing angles. In this way, the energy loss and thus the energy straggling are very small, and the range of angles into which the ions are scattered is very narrow around the angle of specular reflection. The very long and narrow MCP channels restrict the angle range further. In addition, the number of secondary electrons leaving the MCP may already be amplified by a factor that depends on the voltage applied across the MCP.

For a side-by-side comparison of the C-foil and MCP performance in an identical sensor configuration, the C-foil has been replaced by a single-stage MCP in one of the 16 sectors of the TOF section. A $3.5 \mu\text{g}/\text{cm}^2$ C-foil was compared with an MCP, whose channels have a $25 \mu\text{m}$ diameter and a length to diameter ratio $l/d = 20$. Figure 6 shows TOF spectra taken with an Ar^+ beam at energies of 11, 12.7, and 16 keV. The C-foil spectra are shown on the left, and the ones with the MCP on the right. A remarkable reduction in the width of the TOF spectra with the MCP entrance is evident. For 11 keV, the full width at half maximum is reduced by almost a factor of 3 from ≈ 125 to ≈ 45 TOF channels. Because of its high mass, Ar is usually an extremely challenging

ratios on solar wind speed [Kucharek *et al.*, 1998], also confirmed by Kallenbach *et al.* [1998]. A likely reason for this fractionation may be found in the inefficient Coulomb drag in the corona [Bodmer and Bochsler, 1998].

5.3. Improvement of TOF Mass Resolution Without Reflection

While the reflecting TOF scheme appears to be an ideal solution for an isotope-resolving TOF mass spectrograph, it comes with challenges and shortcomings. First, the reflecting electric field requires an additional high-voltage supply for rather high voltages to cover a reasonable energy range. Second, the retarding field naturally defocuses the ion trajectories. As a consequence, the already existing angular dispersion after the entrance foil increases substantially so that only a small fraction of the incoming ions are detected at the stop detector, thus reducing the effective geometric factor of this spectrograph typically by more than 1 order of magnitude over that of a linear TOF system. Therefore, alternatives to improve TOF resolution needed.

What limits the TOF resolution of a linear TOF spectrograph most severely is the energy and angle straggling during the ion passage through the foil. Therefore, much effort has gone into reducing the foil thickness [Funsten *et al.*, 1994; McComas *et al.*, 2004]. A potential alternative is the suggestion to replace the foil as the secondary electron generating element with a

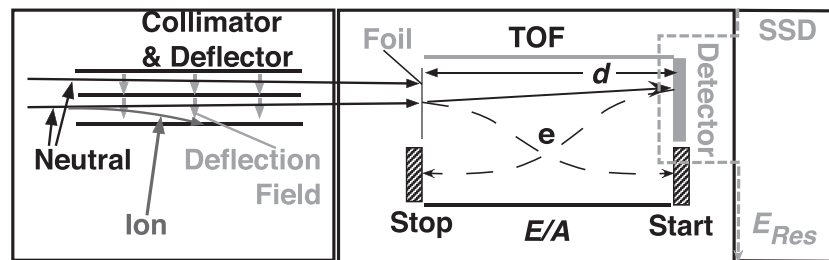


Figure 7. Schematic view of a suprathermal ENA mass spectrograph. (left) A mechanical collimator with electrostatic deflection filters out incoming ions in the ENA energy range. (right) The TOF section is similar as in section 3.

species to obtain a TOF spectrum at such low energies. Building on advanced atomic layer deposition, this test setup is also being used to test the reflectivity, scattering characteristics, and secondary electron yield of several different MCP channel coatings. The finite open area ratio of the MCPs and their limited opening solid angle reduce the transparency of the sensor entrance window and consequently the start efficiency of the TOF section.

6. Energetic Neutral Atom Spectrographs

TOF mass spectrographs work just as well for neutral atoms as they do for ions. Whether a neutral atom or an ion passes through the thin entrance foil does not affect the number of secondary electrons emitted, and the average charge state upon exit from the foil only depends on the energy and species of the particle [Gonin *et al.*, 1992]. Therefore, TOF spectrographs can also be used to analyze energetic neutral atoms (ENAs). Neutral atom populations are ubiquitous in the solar system, in planetary, lunar, and cometary atmospheres, as well as in the InterStellar Neutral (ISN) gas flow through the heliosphere.

The simultaneous presence of plasmas and neutral gas provides for a very effective remote sensing technique of ion velocity distributions in the heliosphere for populations that are found at a large distance from the observer. Because charge exchange occurs through the transfer of an electron from the neutral gas atom, with the electron mass negligible compared to that of the parent ion, the daughter ENA maintains the original ion momentum. Therefore, the ENA velocity distribution mimics that of the parent ion distribution, modified by any speed dependence in the charge exchange cross-section $\sigma_{\text{ChEx}}(v)$. ENAs are not deflected by electric or magnetic fields on their way to the observer and arrive on straight trajectories, only modified by gravitational deflection [e.g., Kucharek *et al.*, 2015], radiation pressure, ionization losses [Bzowski *et al.*, 2013], and collisions. Before corrections for the aforementioned effects, the observed differential ENA fluxes J_{ENA} as function of energy E and solid angles θ and Φ are related to the parent ion velocity distribution $f_i(v, \theta, \Phi)$ according to:

$$\frac{dJ_{\text{ENA}}(E, \theta, \Phi)}{dE d\Omega} = n_a \sigma_{\text{ChEx}}(v) \Delta l \cdot \frac{v^2}{m_i} f_i(v, \theta, \Phi) \quad (9)$$

n_a is the density of the surrounding neutral gas, Δl is the length along the line of sight over which the charge exchange interaction occurs, and m_i is the mass of the ion. The factor v^2/m_i provides the transformation from velocity distribution to differential flux as in equation (1). One of the biggest challenges to making reliable observations of ENAs is the competition from typically much higher fluxes of ions at comparable energies. These ions must be suppressed before entering the sensor, or they must be carefully distinguished inside the sensor system, because the detectors are equally sensitive to ions and neutral atoms. Of course, other background sources for particle measurements, such as photons, electrons, and penetrating cosmic rays, have to be eliminated or at least well characterized.

6.1. Suprathermal Neutral Atom Spectrometers

Because ions and neutral atoms produce secondary electrons at the TOF entrance foil with comparable efficiencies, suprathermal and energetic particle TOF spectrographs without an ESA entrance system, such as SAMPEX LICA or ACE ULEIS [Mason *et al.*, 1998], can also work as ENA mass spectrographs. However, they need an electrostatic suppression scheme at the entrance aperture so that ENAs can be identified unambiguously. The first sensor equipped in this way is the High energy section (HSTOF) section of the SOHO CELIAS Suprathermal TOF (STOF) sensor [Hovestadt *et al.*, 1995]. A schematic view of this sensor is shown in

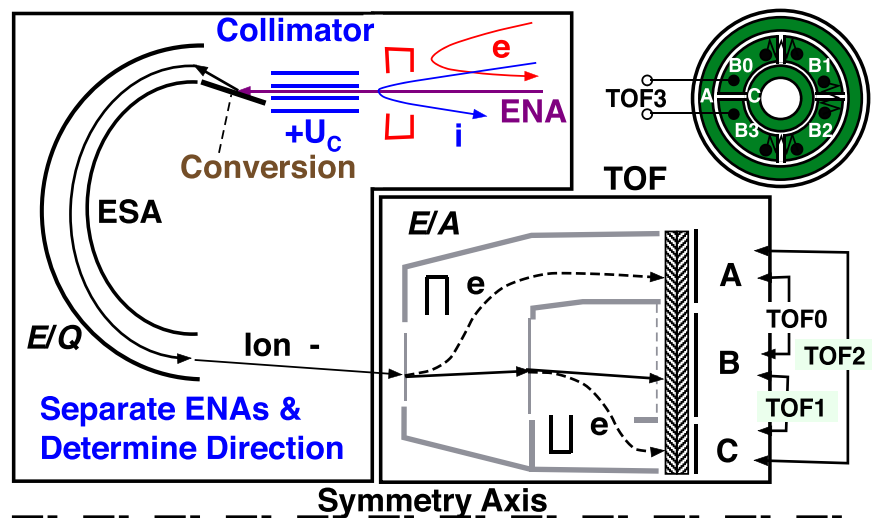


Figure 8. Schematic radial cut of IBEX-Lo, along with its symmetry axis. A top view of the signal anodes is shown in the upper right with the 4 quadrant anodes in the Stop portion (B) of the MCP, which are connected by delay lines. All 4 TOF measurements (TOF0, 1, 2, 3) are indicated.

Figure 7. The entrance collimator consists of a set of plates that deflect ions of energies up to the upper limit of the ENA energy range E_U/Q to hit the plates, and thus they do not enter the sensor system. Ions with energies higher than E_U/Q will enter, and thus this sensor will serve a dual purpose, as ENA sensor for $E_{ENA} < E_U$ and as ion sensor for $E_{ion}/Q > E_U/Q$. HSTOF is the high-energy ion section of the CELIAS STOF sensor, with a substantial overlap with STOF over the energy range 150–600 keV/Q. This overlap allows an additional validation and separation of ion fluxes that might interfere with the ENA spectrum.

These unique capabilities have enabled the HSTOF sensor to obtain heliospheric ENA distributions in the energy range 55–150 keV [Hilchenbach *et al.*, 1998]. They used the fact that the energy spectra of H ENAs and H^+ seen by HSTOF and STOF above 150 keV match to conclude that there is indeed a substantial excess of H ENAs below 150 keV, increasing toward lower energies. This comparison validated that HSTOF indeed sees H ENAs, unambiguously distinguished from interfering backgrounds. The maximum of this ENA flux appears to come from the heliotail direction [Hilchenbach *et al.*, 1998]. In addition, He ENAs have been identified with HSTOF [Czechowski *et al.*, 2004].

While SOHO HSTOF was primarily designed as an energetic ion TOF mass spectrograph and the ENA separation capability was included only during the implementation phase, other ENA detectors were optimized specifically for that purpose. As part of the Imager for Magnetopause-to-Aurora Global Exploration (IMAGE) mission, the High Energy Neutral Atom (HENA) imager [Mitchell *et al.*, 2000] was optimized specifically for the imaging of magnetospheric ENAs, largely using the same physical principles of HSTOF. The HENA sensor provided valuable insight into the global dynamics of energetic particles during geomagnetic storms [e.g., Mitchell *et al.*, 2001]. The Cassini probe carries the Ion-Neutral Camera (INCA) [Mitchell *et al.*, 1996] as part of the magnetosphere imaging instrument package [Krimigis *et al.*, 2004]. INCA was designed and optimized to image Saturn's magnetosphere and its interaction with the moons, but it also provides images of the heliospheric boundary region [Krimigis *et al.*, 2009] through ENAs at energies above the range covered by the Interstellar Boundary Explorer (IBEX) mission [McComas *et al.*, 2009a].

6.2. Low Energy Neutral Atom Spectrographs

In a similar way, low-energy ion TOF mass spectrographs can be used for low-energy neutral atoms as well. The main challenge is that neutral atoms cannot be manipulated with electric and magnetic fields like ions. In particular, an ESA is needed to determine the energy of low-energy particles; and, as discussed in section 4, ions at energies below about 10 keV/Q need to be postaccelerated to obtain a reasonable TOF and thus mass resolution. Therefore, in all low-energy sensors ENAs are converted into ions after they have passed a collimator at the entrance, which allows the determination of their arrival direction.

All low-energy ENA sensors contain the same building blocks as shown in Figure 8. The sensor aperture and electrostatic deflection and/or repeller scheme reject electrons and ions from entering the collimator. A mechanical collimator selects the arrival direction and determines the sensor FoV for the incoming ENAs. Then the ENAs are converted to ions. For medium low energies, typically 0.5–10 keV, an ultrathin C-foil is used to strip off electrons to convert atoms to positive ions. The fraction of neutrals that become ions depends on species and energy [Gonin *et al.*, 1992]. Such a sensor has been built and optimized for magnetospheric imaging for IMAGE as the Medium Energy Neutral Atom imager [Pollock *et al.*, 2000]. For ENA energies between 0.01 and 2 keV, the ENAs undergo a reflection at glancing incidence angles on a highly polished surface and are converted to negative ions, as implemented for IMAGE in the Low Energy Neutral Atom (LENA) imager [Moore *et al.*, 2000] and developed further in Livi *et al.* [2003]. Figure 8 shows a configuration of the low-energy version of the ENA sensor with reflection off a conversion surface. After conversion, the emerging ions pass an ESA for energy selection and then are postaccelerated. The final functional block is a TOF mass spectrograph or, if mass analysis is not needed, a coincidence detector system. The implementation of a coincidence system becomes very important for ENA cameras because the expected ENA fluxes are generally extremely low compared with those of ions in the same energy range and also compared to any background generating signals. Therefore, the coincidence of two or more related signals that are generated by the incoming ENAs is required to distinguish the weak signal from any kind of background. Low-energy ENA imagers that employ surface conversion to positive ions and the subsequent bounce on a surface with secondary electron generation as start signal are used for planetary applications [e.g., Barabash *et al.*, 2006, 2009]. Here we concentrate on heliospheric applications with their background related challenges.

IBEX takes ENA imaging to its current limits by obtaining images of the heliosphere boundary and sampling directly the neutral interstellar gas flow through the inner heliosphere [McComas *et al.*, 2009a]. To achieve these goals, particular attention to background and noise suppression was necessary at all steps, as mentioned above [Wurz *et al.*, 2009], because the signal rates that make up the ENA sky maps are extremely low, so that some individual pixels of the maps contain only a few ENA counts per day. Therefore, the IBEX detector units are equipped with genuine triple coincidence systems. IBEX contains two cameras, IBEX-Lo for the low-energy range 0.01–2 keV [Fuselier *et al.*, 2009] with ENA to negative ion conversion and IBEX-Hi for the medium low-energy range 0.3–6 keV [Funsten *et al.*, 2009] with stripping of ENAs to positive ions. Here we will concentrate on the IBEX-Lo TOF spectrograph.

The IBEX-Lo TOF system is adapted from Cluster CODIF [Rème *et al.*, 1997]. Because background suppression is so important for IBEX, two similar TOF sections are nested to provide genuine triple coincidence. Figure 8 shows a schematic radial cut. At the first C-foil, each incoming ion generates secondary electrons that are guided to the outermost section (A) of a single two-stage MCP, while electrons from the second foil reach the innermost section (C). The ions are stopped at the center section, which consists of four quadrants (B0–B3) for additional background characterization. All sections are electrically separated on the signal anode (Figure 8, top right). TOF measurements are made between each of the foils and the MCP (TOF0 and TOF1) and between the two foils (TOF2). For additional background characterization, the quadrant position is derived from a delay line along B0–B3 (TOF3).

During the IBEX-Lo background characterization each of the three TOF coincidence rates was lower than 10^{-3} s^{-1} , indicating a very clean and low-noise TOF section. Next we turn to the advantages of a triple TOF configuration to reduce chance signal coincidences due to orphaned start and stop signals, as already indicated in equations (4) and (4a). The additional TOF signal reduces the background rate by another factor $R_{\text{start}} \Delta \tau$, where $\Delta \tau$ is the TOF resolution. Figure 9 shows how requiring more TOF signals and successive constraints improve the background response with TOF2 (H1 and H2) and TOF0 (H3 and H4) histograms and TOF1 versus TOF0 scatterplots (S1–S4) from an H atom test with 16 kV postacceleration. S1, H1, and H3 contain all events with at least one of the three TOF measurements valid. Except for an easily identifiable peak near 0 ns in TOF2 (related to electrons) and two very small “ghosts” (also in S1) near 0 ns in TOF0, the single TOF histograms provide already a clean H distribution [Möbius *et al.*, 2008].

S2 and H2 only include events with four valid TOF measurements and which fulfill the checksum condition: $|\text{TOF0} + \text{TOF3} - \text{TOF1} - \text{TOF2}| < 1 \text{ ns}$. Clearly, all events with $\text{TOF} = 0$ in one of the channels (no valid TOF) are gone and most of the events outside the H peak region. In H3, S1, and S2 two distinct peaks (connected by arrows) are seen, which come from neighboring MCP quadrants with different delay times. In S3, S4, and

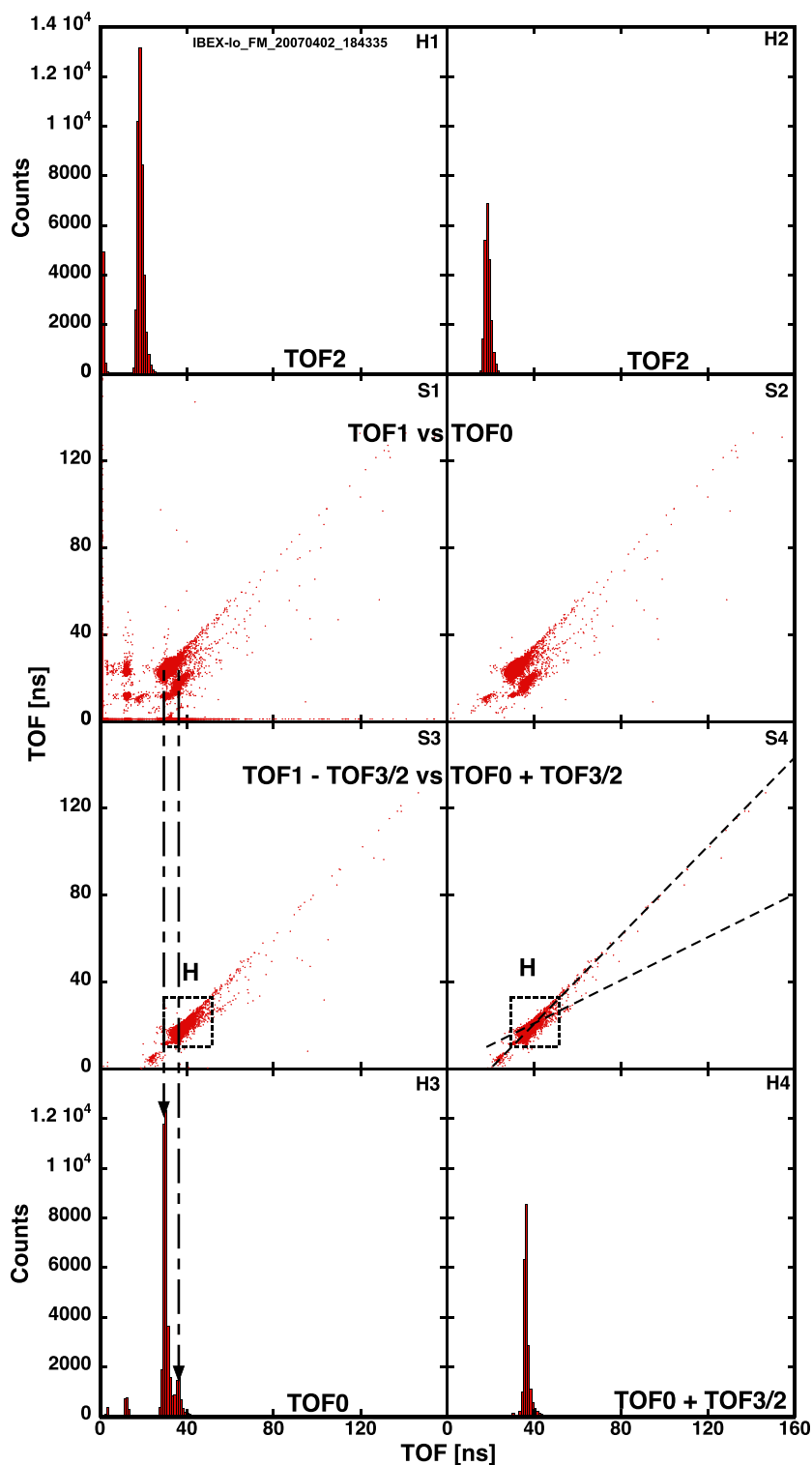


Figure 9. Sequence of TOF2 histograms H1 and H2, TOF1 versus TOF0 scatterplots that shows the successive removal of background and ghost signals from S1 through S4, and TOF0 histograms H3 and H4, as described in the text (adapted from Möbius *et al.* [2008]; reproduced with the permission of the Institute of Geophysics-UNAM).

H4 this is corrected by using TOF0 + TOF3/2 and TOF1 – TOF3/2 instead, perfectly aligning the H distribution. Only events inside the (dashed) H box remain and a few along two diagonal lines with slopes of 1 and 0.5 (faint dashed lines in S4), due to H energy straggling in the second or first foil, respectively. S3 and S4 show a residual <0.5% peak on the slope 1 line, which stems from cross talk to the delay line anode and thus also contains legitimate H events. Like the H ENA selection in the flight data, in S4 only events have been selected in TOF2 (H2) that belong to the H peak down to the 1% level, thus requiring consistency between all three TOFs. During a background run over 2.5 days no triple H count was recorded, thus exceeding even our ambitious goal of less than 1 count per day.

In spite of its emphasis on excellent background suppression, the IBEX-Lo TOF subsystem also provides high detection efficiency, which is very homogenous over the entire unit. For highest background suppression triple coincidence is used, and for highest counting statistics any single TOF measurement is accepted. Both efficiencies have been derived from calibrations with neutral beams for the nominal postacceleration voltage. Even with a postacceleration voltage that is now reduced to 7 kV from 16 kV after a discharge during a long eclipse, IBEX-Lo works very effectively. It still fulfills all science objectives, except for observations of the O and Ne interstellar gas flow. The TOF efficiencies for H ENAs and the sputtered H component from the neutral interstellar He flow are reduced by about a factor of 2, and the background is slightly elevated, but these changes do not impact the ENA observations significantly. In other words, IBEX-Lo still takes very clean observations.

6.3. Heliospheric Low-Energy ENA Observations

The excellent noise and background suppression, good mass resolution, and large collecting power of IBEX-Lo have made several important heliospheric discoveries possible, including the detection of the IBEX ribbon at low energies [McComas *et al.*, 2009b]. During the implementation phase of IBEX it was not evident whether the H ISN flow would be detectable at 1 AU because of its very high ionization losses and the generally higher background susceptibility at the low H energies of 10–30 eV. Following the discovery of the He ISN flow with Ulysses GAS [Witte *et al.*, 1996] and observations with IMAGE LENA [Collier *et al.*, 2004], IBEX-Lo clearly identified the H ISN flow, along with the expected He and O ISN signal [Möbius *et al.*, 2009]. For the first time, the effect of solar radiation pressure on the H ISN flow could be deduced quantitatively from direct particle observations [Schwadron *et al.*, 2013].

By comparing the observed ratio of sputtered C and converted O in the O ISN flow signal with calibration results using an O and a Ne beam, it was even possible to identify a Ne component in the ISN flow and to determine the Ne/O ratio of the interstellar gas outside the solar system [Bochsler *et al.*, 2012; Park *et al.*, 2014]. The resulting ratio for the pristine interstellar medium agrees with findings from pickup ion observations by Gloeckler and Geiss [1998], but these values suggest a higher Ne/O ratio than found in the solar system. This raises the question whether there are genuinely different abundances in the solar system compared with the local neighborhood or whether a substantial amount O may be locked in interstellar dust.

Most recently, IBEX-Lo has revealed secondary ISN flow components for O [Möbius *et al.*, 2009; Park *et al.*, 2015] and He [Kubiak *et al.*, 2014]. These are thought to arise from charge exchange between the diverted interstellar plasma in the outer heliosheath and the neutral interstellar flow. Such a component was predicted for O, but thought to be very weak for He [Müller and Zank, 2004]. Their characteristics will shed light on the interaction process in the outer heliosheath and on the plasma diversion caused by the interstellar magnetic field.

7. Conclusions and Outlook

TOF spectrographs for space applications have come a long way since their first use on satellites in the mid-1980s. They can be adapted to a variety of uses and parameter ranges by combining several different building blocks and optimizing them based on prior experience and new needs. The inclusion of novel technologies, such as the potential replacement of C-foils by a single-stage MCP to achieve better TOF resolution, advanced ESA concepts to improve the effective geometric factor and duty cycle, as well as the use of modern solid state detector concepts to combine timing and energy measurement, ensures that the reach of TOF instrumentation will go even further. Large-scale integration technologies reduce the resource requirements of the electronics, and modern processor technologies help to optimize the data stream and to further reduce resources so that the electronics for TOF instruments can be reduced in size. The actual sensors are scaled based on the required collection capability, the required TOF path length, and high-voltage constraints, so these aspects are much

harder to miniaturize. TOF instruments have a wide range of applications from ionospheric and magnetospheric physics, via solar, interplanetary and interstellar physics to planetary applications.

With their tremendous achievements in the IBEX mission, TOF spectrographs also have paved the way to a more ambitious follow-on mission, the Interstellar Mapping and Acceleration Probe (IMAP). This mission, when implemented, will bring the ENA mapping of the heliosphere boundary and the sampling of the interstellar gas flow to the next level of resolution and collecting power. It will combine this capability with the crucial pickup ion diagnostics of the interstellar gas population and high cadence observations of suprathermal and energetic particles to trace their acceleration from the parent population to high energies. IMAP has been promoted as the highest priority for solar terrestrial probes in the recent Heliophysics Decadal Survey [Baker et al., 2013]. The scientific objectives of this mission depend crucially on a complement of TOF spectrographs, which are optimized for neutral atoms over an extended energy range compared with IBEX and for ions from pickup ion energies to about 1 MeV/nuc. The ENA TOF spectrographs can be optimized based on ample test, calibration, and flight experience with IBEX and Cassini, and the PUI and suprathermal sensor will be the first dedicated instrument for this purpose based on ample experience from Ulysses, ACE, and STEREO. Without the advances made over the past 30 years in TOF technology and supporting sensor modules such mission would not be possible.

Acknowledgments

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Erratum

In Figure 8 of this article, the time-of-flight section (lower right portion) was accidentally flipped upside down compared with the actual sensor configuration. This error has been corrected, and a line indicating the rotational symmetry axis of the sensor has been added for clarity. This may be considered the authoritative version of record